

Heart Failure Prediction Dataset

Exploratory Data Analysis — Key Insights & Recommendations

918

Patient Records

12

Clinical Features

55.3%

Heart Disease Rate

0

Missing / Duplicate Rows

Executive Summary

This report summarizes exploratory data analysis (EDA) performed on the Heart Failure Prediction dataset (918 patient records, 12 clinical attributes). The dataset is clean — no missing values or duplicate rows were found — and is moderately imbalanced by sex (79% male, 21% female) but well balanced by outcome (55.3% positive for heart disease). Four clinical variables stand out as the strongest signals for heart disease risk: **maximum heart rate (MaxHR)**, **ST depression (Oldpeak)**, **fasting blood sugar**, and **age**. Asymptomatic chest pain (ASY) is the single most common presentation and is disproportionately associated with a positive diagnosis, an important and clinically consistent finding since asymptomatic patients often present later and with more advanced disease.

Dataset Overview

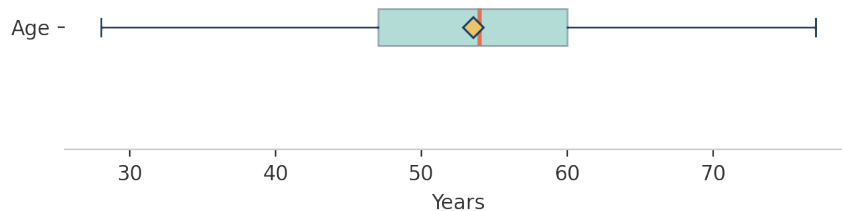
The dataset contains 918 rows and 12 columns spanning demographic (Age, Sex), clinical measurement (RestingBP, Cholesterol, FastingBS, MaxHR, Oldpeak), diagnostic (ChestPainType, RestingECG, ExerciseAngina, ST_Slope), and outcome (HeartDisease) fields. During preprocessing, the analysis confirmed zero null values across all columns, zero duplicate records, and renamed the 'Sex' column to 'Gender' for clarity. No rows required removal.

Metric	Value
Age (years)	Mean 53.5 Median 54 Range 28 - 77
Resting Blood Pressure (mm Hg)	Mean 132.4 Median 130 Range 0 - 200
Cholesterol (mg/dl)	Mean 198.8 Median 223 Range 0 - 603
Max Heart Rate (bpm)	Mean 136.8 Median 138 Range 60 - 202
Oldpeak (ST depression)	Mean 0.89 Median 0.6 Range -2.6 - 6.2
Fasting Blood Sugar > 120 mg/dl	23.3% of patients

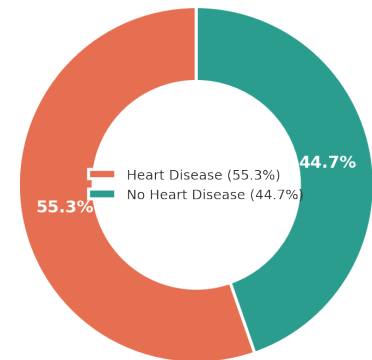
Univariate Analysis

Age Distribution

Age Distribution Summary (n=918)



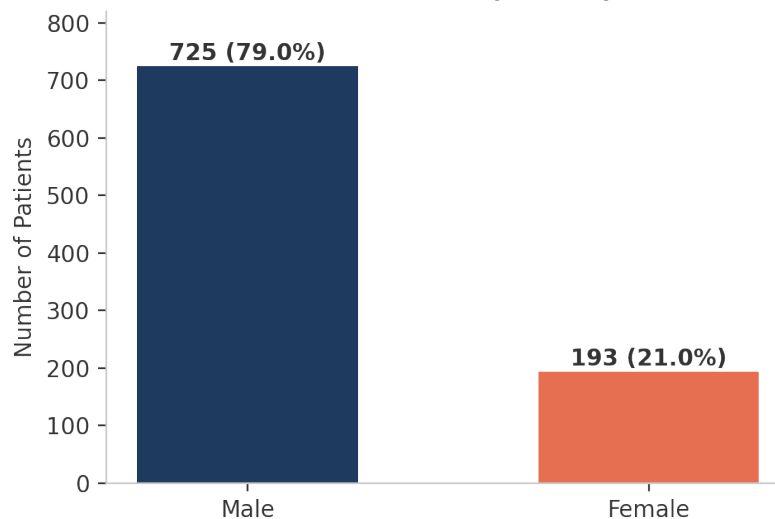
Heart Disease Prevalence in Dataset



Patients skew middle-aged to older adult, with an interquartile range of 47-60 years and a median of 54. The distribution is roughly symmetric with a slight concentration in the 50s, consistent with the population most at risk of coronary disease. The dataset's overall heart disease prevalence of 55.3% makes it well-suited for classification modeling — close to a balanced target.

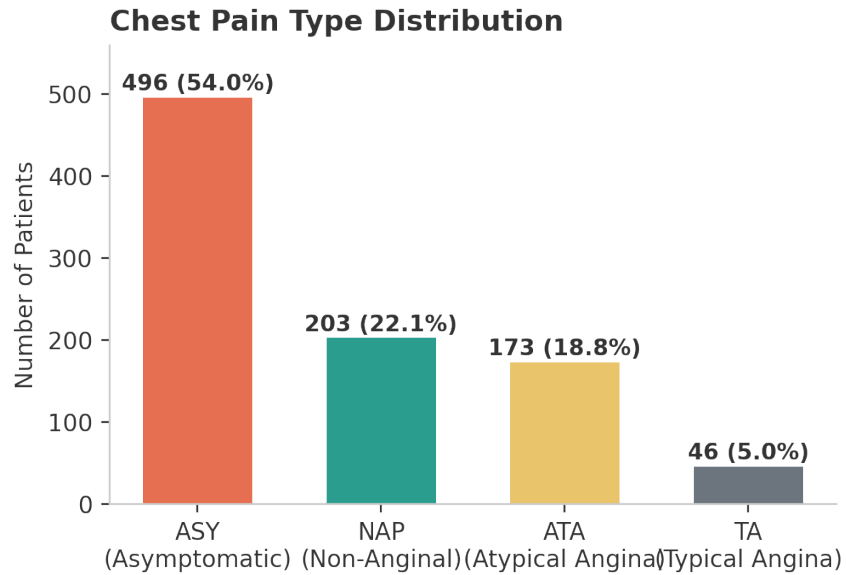
Sex Distribution

Patient Sex Distribution (n=918)



Men make up 79.0% (725) of the sample versus 21.0% (193) women — a substantial imbalance that should be accounted for when generalizing findings or training predictive models, since risk patterns and the reliability of estimates may differ by sex, and results skew toward male physiology.

Chest Pain Type Distribution



Asymptomatic chest pain (ASY) is the most frequent category at 54.0% (496 patients), more than double the next most common type (NAP, non-anginal, 22.1%). This is a clinically notable pattern: patients who are asymptomatic often go undiagnosed longer, and — as shown in the bivariate analysis below — ASY patients in this dataset carry a disproportionately high rate of confirmed heart disease.

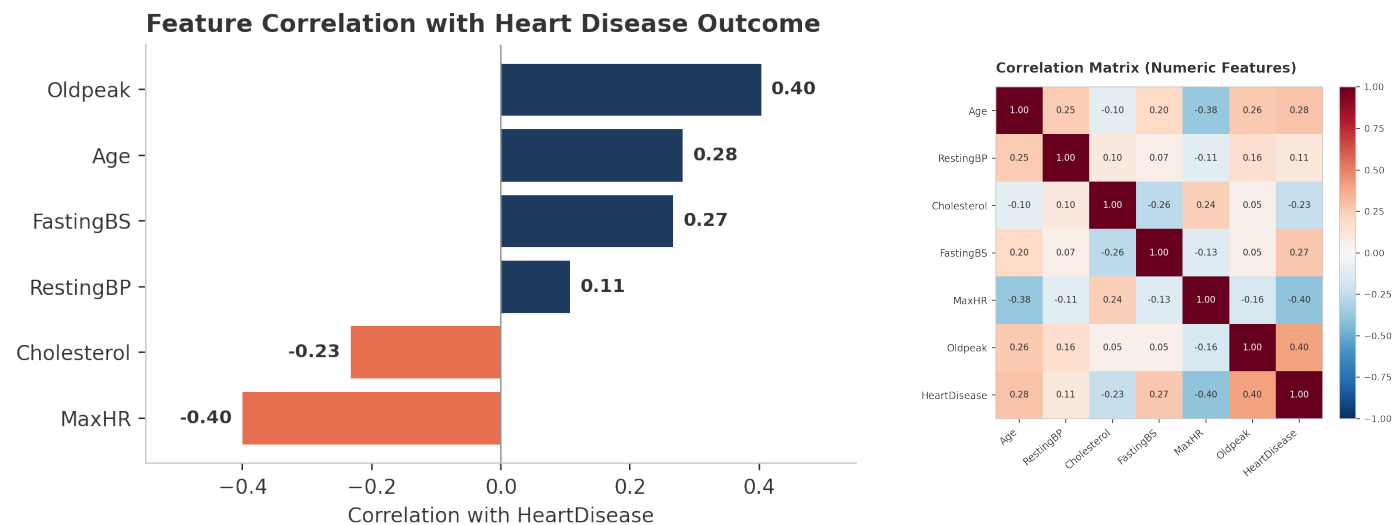
Bivariate & Multivariate Analysis

Chest Pain Type × Sex × Age

Chest Pain Type	Female — Avg Age	Male — Avg Age	Max Age Observed
ASY (Asymptomatic)	54.3	55.1	77
ATA (Atypical Angina)	49.6	49.1	74
NAP (Non-Anginal Pain)	53.1	53.4	76
TA (Typical Angina)	54.2	55.0	74

Average age is remarkably consistent across chest pain types and sexes (49-55 years), with ATA (atypical angina) patients being the youngest on average. Within every chest pain category, the most frequent sex/type combination overall is **ASY presenting in males**, the single largest at-risk subgroup in the dataset.

Correlation with Heart Disease Outcome



- **Oldpeak (ST depression, $r = 0.40$)** and **Age ($r = 0.28$)** are the strongest positive correlates — higher ST depression during exercise and older age both track with higher heart disease likelihood.
- **MaxHR ($r = -0.40$)** is the strongest negative correlate — patients who cannot reach a high maximum heart rate under exertion are more likely to have heart disease, a classic reduced cardiac-reserve signal.
- **Fasting blood sugar ($r = 0.27$)** and **Cholesterol ($r = -0.23$)** also show meaningful — though weaker — relationships; the negative cholesterol correlation is counter-intuitive and likely reflects the zero-value data quality issue noted below rather than a true protective effect.
- **Resting blood pressure ($r = 0.11$)** shows only a weak standalone relationship with the outcome in this dataset.

Data Quality Notes

- No missing values (NaNs) were detected in any of the 918 rows across 12 columns, and no duplicate records were found — the dataset required no row-level cleaning.
- Both **Cholesterol** and **RestingBP** contain a minimum value of 0, which is not physiologically plausible and almost certainly represents unrecorded/missing measurements encoded as zero rather than true readings. These zeros likely distort the mean, standard deviation, and correlation estimates for Cholesterol in particular — treating them as missing (e.g., via imputation) is recommended before using the feature in any predictive model.
- The 'Sex' column was renamed to 'Gender' during preprocessing; downstream analysis or modeling should confirm consistent naming across any additional data sources.
- Class balance (55.3% positive) is favorable for building a classification model without needing resampling techniques.

Key Insights Summary

- The dataset is clean and well-balanced by outcome (55.3% heart disease) but skewed by sex (79% male) — sex-stratified analysis is advisable before drawing population-wide conclusions.
- Asymptomatic chest pain (ASY) is both the most common presentation (54%) and the type most associated with confirmed heart disease, particularly among men.
- MaxHR and Oldpeak are the two strongest individual predictors of heart disease in this dataset ($|r| \approx 0.40$) and should be prioritized as core features in any predictive model.
- Age and fasting blood sugar add incremental predictive signal; resting blood pressure alone is a weak predictor.
- Zero-value entries in Cholesterol and RestingBP need remediation (imputation or exclusion) before the data is used for modeling, as they currently understate true clinical risk relationships.

Recommendations

- **Clean the zero-value readings:** impute or flag the 0 mg/dl Cholesterol and 0 mm Hg RestingBP entries as missing rather than valid measurements before further analysis or modeling.
- **Prioritize MaxHR, Oldpeak, Age, and ChestPainType** as leading features in a predictive model — they show the clearest, most clinically coherent relationships with the outcome.
- **Collect or seek additional female patient records** to reduce the sex imbalance and improve the reliability of conclusions for women.
- **Investigate the ASY-heart disease relationship further** (e.g., via logistic regression or odds ratios) given its high prevalence and apparent association with positive diagnoses.
- **Proceed to model building** (e.g., logistic regression, random forest, or gradient boosting) now that the data has been validated as free of nulls/duplicates — the balanced target class means minimal preprocessing is needed for class imbalance.

End of report.